

I. PROJECT OVERVIEW

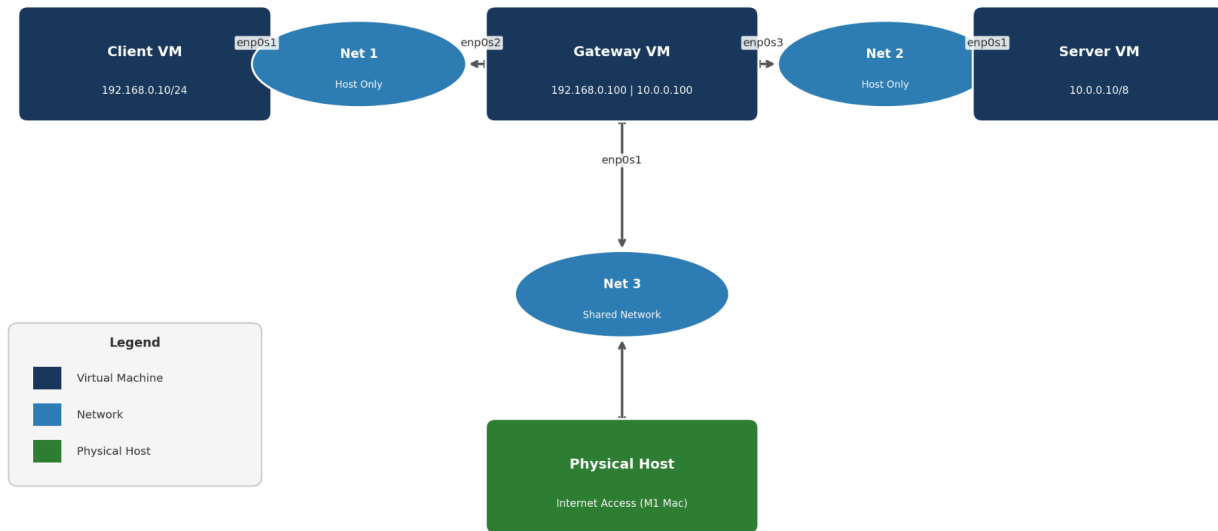
This project involved setting up a virtual network environment using UTM (a hypervisor for Apple Silicon) and Ubuntu 22.04.1 LTS ARM64 virtual machines on an M1 Mac. Three VMs were created and configured: a Client VM, a Gateway VM, and a Server VM. The Gateway VM was connected to three networks: an internal client-side network (Net1), an internal server-side network (Net2), and a Shared Network providing Internet access (Net3). Network interfaces on each VM were configured manually using the `ip` command and permanently via `netplan` configuration files. IP forwarding and NAT masquerading were enabled on the Gateway to allow the Client and Server VMs to route traffic to the public Internet through the Gateway. Connectivity was verified using ping tests to confirm all three VMs could reach each other and external hosts such as `google.com`.

II. NETWORK SETUP

Draw a diagram and provide descriptions about the network setup, including:

- Topology,
- Network configurations (IP addresses, etc.)
- Initial reachability among network nodes

Network Topology — Project 01



Network Configurations

VM	Interface	IP Address	Network / Role
Client VM	enp0s1	192.168.0.10/24	Net1 - Internal client network; default gateway: 192.168.0.100
Gateway VM	enp0s2	192.168.0.100/24	Net1 - Client-facing internal network interface
Gateway VM	enp0s3	10.0.0.100/8	Net2 - Server-facing internal network interface
Gateway VM	enp0s1	DHCP assigned	Net3 - NAT Network (Internet access via host)
Server VM	enp0s1	10.0.0.10/8	Net2 - Internal server network; default gateway: 10.0.0.100

DNS Servers

All VMs were configured with the following DNS nameservers: 8.8.8.8 (Google Primary) and 8.8.4.4 (Google Secondary).

Initial Reachability

After completing all network configurations, reachability was verified with the following ping tests:

- [1] From Client VM: ping 192.168.0.100 (Gateway - Net1 interface) - Success
- [2] From Client VM: ping 10.0.0.10 (Server VM) - Success (routed via Gateway)
- [3] From Client VM: ping google.com - Success (Internet via Gateway NAT)
- [4] From Server VM: ping 10.0.0.100 (Gateway - Net2 interface) - Success
- [5] From Server VM: ping google.com - Success (Internet via Gateway NAT)
- [6] From Gateway VM: ping google.com - Success

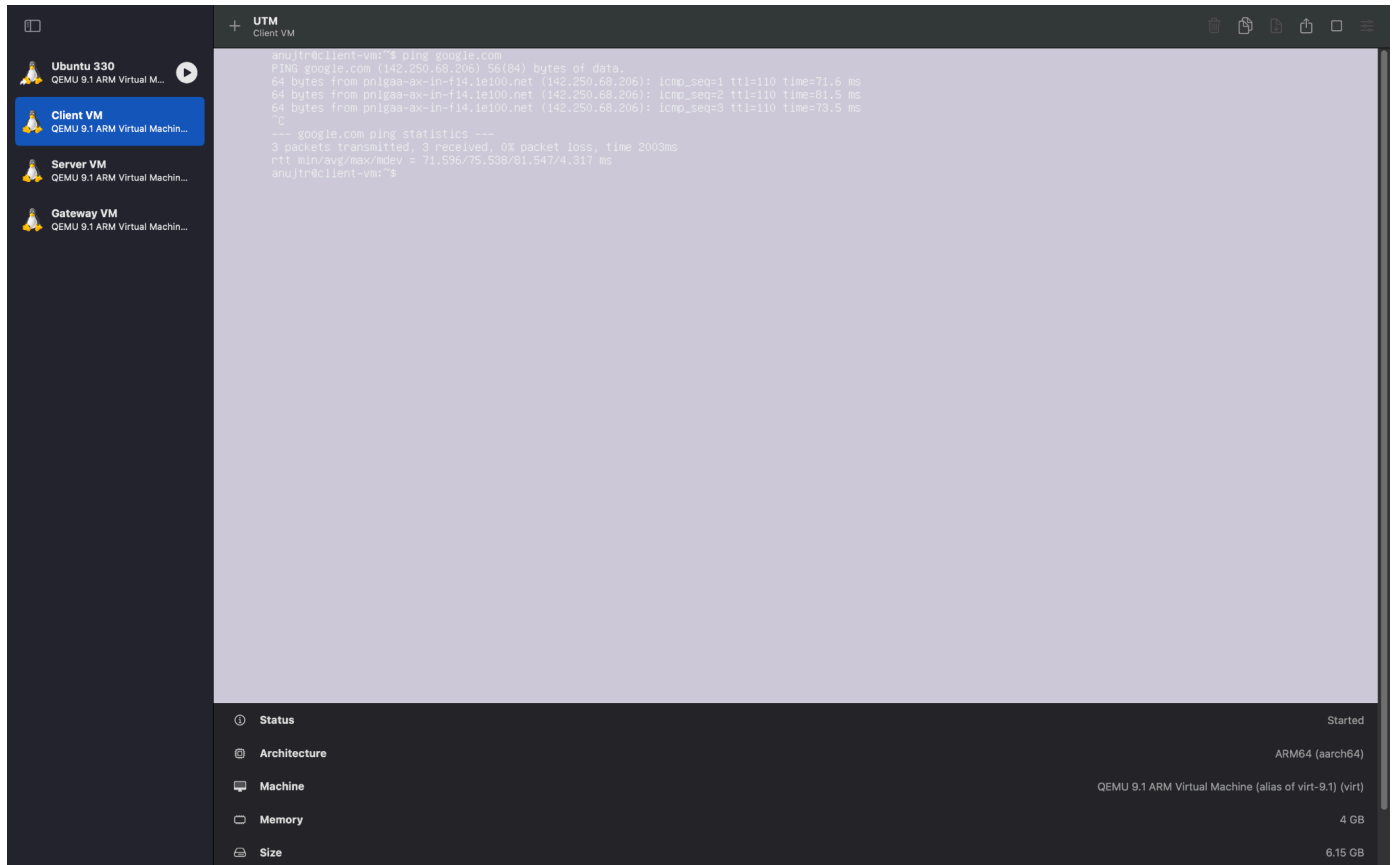
III. SOFTWARE

Software	Purpose
UTM 4.x (Apple Silicon)	Hypervisor for Apple Silicon (M1/M2/M3) used to create and manage the three virtual machines. UTM uses QEMU under the hood and supports ARM64 virtualization natively.
Ubuntu 22.04 LTS	Guest OS installed on all three VMs. Provides the Linux environment for network configuration.
ip (iproute2)	Used to check and assign IP addresses on each VM (ifconfig was unavailable; ip addr and ip link were used instead).
route / ip	Used to add default gateway routes on Client and Server VMs.
netplan	Used to permanently configure network interfaces via YAML configuration files in /etc/netplan/.
dhclient	Used on the Gateway VM to obtain a DHCP-assigned IP address for the NatNetwork (enp0s1) interface.
iptables	Used on the Gateway VM to configure NAT POSTROUTING MASQUERADE rule, enabling Client and Server traffic to reach the Internet.
sysctl	Used to enable IPv4 packet forwarding on the Gateway VM via net.ipv4.ip_forward=1.
ping	Used to verify network connectivity between VMs and to the public Internet (google.com).

IV. PROJECT DESCRIPTION

Task 1: Download and Install UTM

- [7] Navigate to <https://mac.getutm.app> and download UTM for macOS. UTM is required for Apple Silicon (M1/M2/M3) Macs as VirtualBox does not fully support ARM64.
- [8] Open the downloaded .dmg file and drag UTM to the Applications folder.
- [9] After installation completes, launch UTM from the Applications folder.



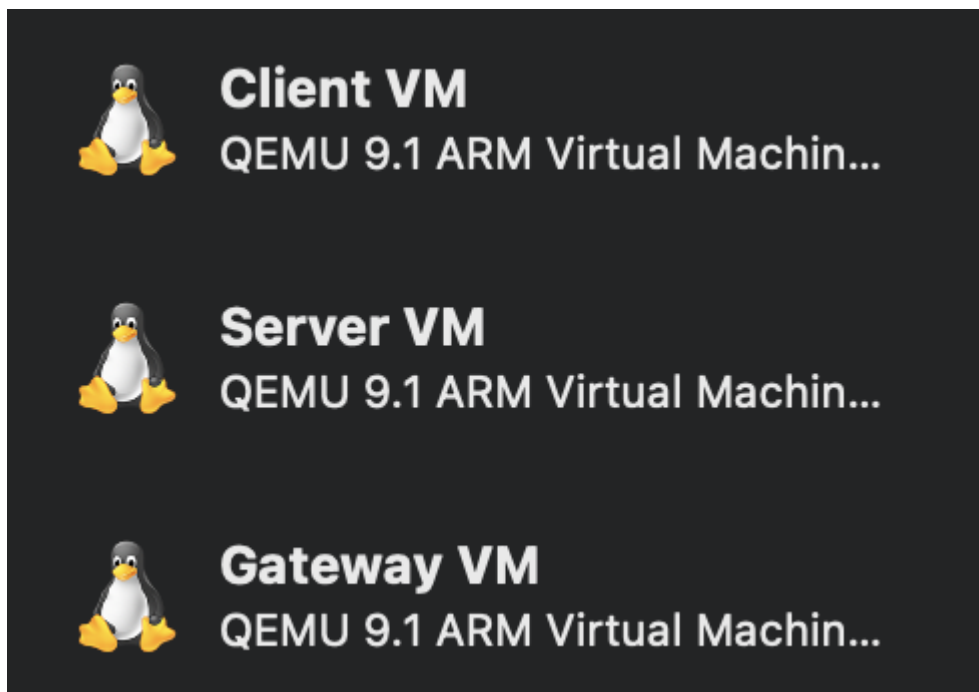
Task 2: Install Ubuntu VM on UTM

The following steps were repeated three times to create the Client VM, Gateway VM, and Server VM (or the first VM was cloned twice).

- [10] Download the Ubuntu 22.04.1 LTS Server ARM64 ISO from <https://cdimage.ubuntu.com/releases/22.04.1/release/> - specifically `ubuntu-22.04.1-live-server-arm64.iso`. The ARM64 server ISO is required for Apple Silicon.
- [11] In UTM, click the + button, select Virtualize, then select Linux.
- [12] Browse to the downloaded ARM64 ISO file. Leave "Boot from kernel image" unchecked.
- [13] On the Unattended Guest OS Install Setup page, set credentials:
 Set hardware resources: at least 4096 MB RAM and 2 CPU cores.
 Set disk size to at least 20 GB. Name the VM (e.g., Client VM) and click Save.
 Start the VM and follow the Ubuntu Server installer. Set hostname (no spaces), username, and password.
 After install completes, stop the VM, remove the ISO from USB Drive in VM settings, then restart.

VirtualBox Only:

- [14] Allocate resources: at least 4GB RAM and 2 CPUs. Keep memory within the green zone to leave resources for the host OS.
- [15] Click Finish. The unattended installation begins automatically. Do not interact with the "Try and Install Ubuntu" prompt.
- [16] After the VM reboots and the Ubuntu login screen appears, log in with the credentials set above.



Task 3: Network Setup in UTM

Network adapters were configured by right-clicking each VM in UTM and selecting Edit, then configuring the Network section. Additional adapters were added using the New button in the Devices section.

Gateway VM - Three Network Adapters:

- [17] Adapter 1 (enp0s1): Network Mode: Shared Network (NAT - Internet, Net3)
- [18] Adapter 2 (enp0s2): Network Mode: Host Only (Net1 - Client network)
- [19] Adapter 3 (enp0s3): Network Mode: Host Only (Net2 - Server network)

UTM's Shared Network mode acts as NAT and provides internet access through the host. No additional setup is required.

Client VM - One Network Adapter:

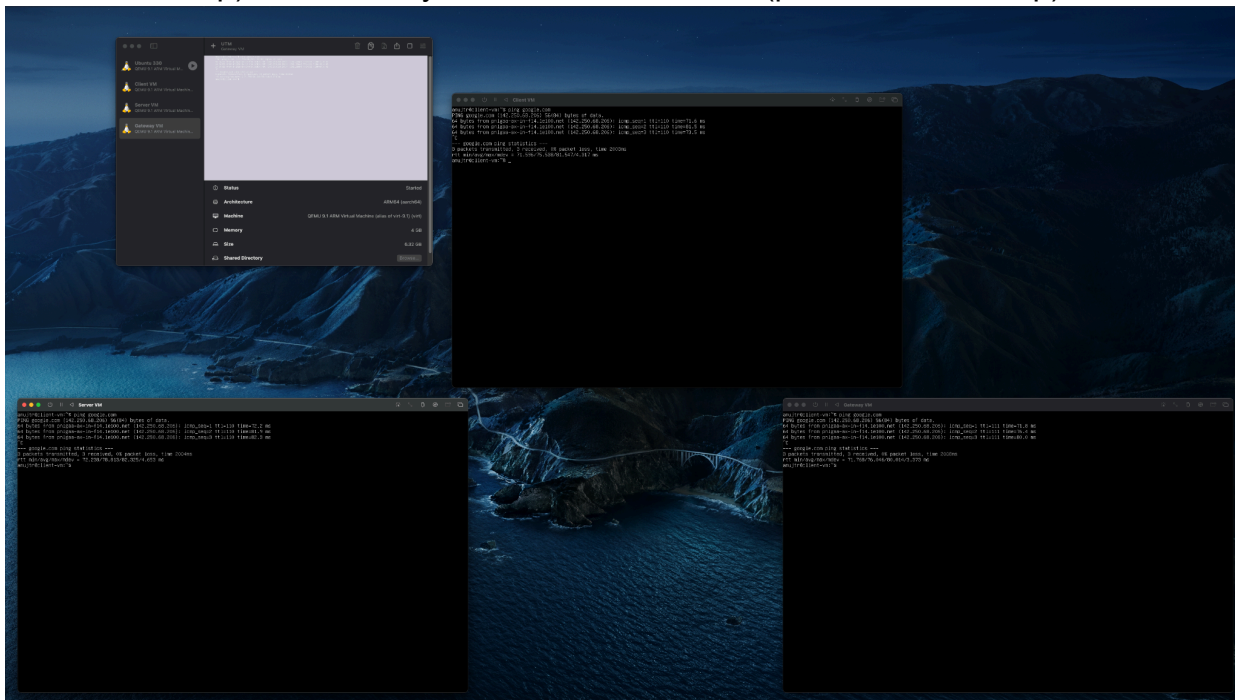
- [20] Adapter 1 (enp0s1): Network Mode: Host Only (Net1 - must match Gateway's Net1 adapter)

Server VM - One Network Adapter:

- [21] Adapter 1 (enp0s1): Network Mode: Host Only (Net2 - must match Gateway's Net2 adapter)

Task 4: VM Operations

- [22] Start all three VMs using the Play button in UTM. Verify each boots to a login prompt.
- [23] Take a snapshot of each VM before making network changes: select VM > Snapshots > Take. This allows rollback if something goes wrong.
- [24] To stop a VM without losing state, use the Pause button in the UTM toolbar (preferred over Stop). Alternatively, use the Pause button (preferred over Stop).



Task 5: Configure VM Networks

Task 5.1: Client VM Configuration

[25] Log into Client VM and open a terminal.

[26] Check available network interfaces:

```
$ ip addr
```

```
anujtr@client-vm:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 12:bc:2f:64:d0:7c brd ff:ff:ff:ff:ff:ff
    inet 192.168.0.10/24 brd 192.168.0.255 scope global enp0s1
        valid_lft forever preferred_lft forever
    inet6 fe80::10bc:2fff:fe64:d07c/64 scope link
        valid_lft forever preferred_lft forever
```

[27] Manually assign IP address:

```
$ sudo ip addr add 192.168.0.10/24 dev enp0s1$ sudo ip link set enp0s1 up
```

[28] Add default gateway route:

```
$ sudo route add default gw 192.168.0.100
# or: $ sudo ip route add default via 192.168.0.100
```

[29] Verify the route:

```
$ ip route
```

```
anujtr@client-vm:~$ ip route
default via 192.168.0.100 dev enp0s1 proto static
192.168.0.0/24 dev enp0s1 proto kernel scope link src 192.168.0.10
```

[30] Make the configuration permanent via netplan:

```
$ sudo vim /etc/netplan/00-installer-config.yaml
```

Netplan configuration file content for Client VM (see Appendix B).

[31] Apply netplan settings:

```
$ sudo netplan apply
```

Task 5.2: Server VM Configuration

[32] Log into Server VM and open a terminal.

[33] Check interfaces and manually assign IP:

```
$ ip addr -a
$ sudo ip addr add 10.0.0.10/8 dev enp0s1$ sudo ip link set enp0s1 up
```

[34] Add default route:

```
$ sudo route add default gw 10.0.0.100
```

[35] Make permanent via netplan (see Appendix B for config):

```
$ sudo vim /etc/netplan/00-installer-config.yaml
$ sudo netplan apply
```

```
anujtr@client-vm:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 1a:09:4e:ba:02:17 brd ff:ff:ff:ff:ff:ff
    inet 10.0.0.10/8 brd 10.255.255.255 scope global enp0s1
        valid_lft forever preferred_lft forever
    inet6 fe80::1809:4eff:feba:217/64 scope link
        valid_lft forever preferred_lft forever
```

Task 5.3: Gateway VM Configuration

- [36] Log into Gateway VM and open a terminal.
- [37] Check available interfaces and assign IPs to Net1 and Net2:

```
$ ip addr
$ sudo ip addr add 192.168.0.100/24 dev enp0s2$ sudo ip link set enp0s2 up
$ sudo ip addr add 10.0.0.100/8 dev enp0s3$ sudo ip link set enp0s3 up
```

- [38] Obtain a DHCP IP address for the NAT network interface (Net3):

```
# enp0s1 obtains DHCP automatically via Shared Network - no manual dhclient needed
```

- [39] Verify all interfaces and the routing table:

```
$ ip addr
$ ip route
```

```
anujtr@client-vm:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 0a:37:3d:d9:f2:22 brd ff:ff:ff:ff:ff:ff
    inet 192.168.64.6/24 metric 100 brd 192.168.64.255 scope global dynamic enp0s1
        valid_lft 3361sec preferred_lft 3361sec
    inet6 fd96:4d82:c6e1:6a82:837:3dff:fed9:f222/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 2591987sec preferred_lft 604787sec
    inet6 fe80::837:3dff:fed9:f222/64 scope link
        valid_lft forever preferred_lft forever
3: enp0s2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 22:01:78:49:5c:46 brd ff:ff:ff:ff:ff:ff
    inet 192.168.0.100/24 brd 192.168.0.255 scope global enp0s2
        valid_lft forever preferred_lft forever
    inet6 fe80::2001:78ff:fe49:5c46/64 scope link
        valid_lft forever preferred_lft forever
4: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 9a:23:00:5f:ca:d9 brd ff:ff:ff:ff:ff:ff
    inet 10.0.0.100/8 brd 10.255.255.255 scope global enp0s3
        valid_lft forever preferred_lft forever
    inet6 fe80::9823:ff:fe5f:cad9/64 scope link
        valid_lft forever preferred_lft forever
```

- [40] Make configuration permanent via netplan (see Appendix B):

```
$ sudo vim /etc/netplan/00-installer-config.yaml
$ sudo netplan apply
```

- [41] Configure iptables NAT rule for Internet forwarding:

```
$ iptables -t nat -A POSTROUTING -o enp0s1 -j MASQUERADE
```

- [42] Enable IPv4 packet forwarding persistently:

```
$ sudo vim /etc/sysctl.conf
Add line: net.ipv4.ip_forward = 1
$ sudo sysctl -p
```

```
anujtr@client-vm:~$ sudo sysctl -p
[sudo] password for anujtr:
net.ipv4.ip_forward = 1
```

Connectivity Verification

After completing all configurations, connectivity was verified from each VM:

```
# From Client VM:
$ ping google.com
```

```
# From Server VM:
$ ping google.com
```

```
# From Gateway VM:
$ ping google.com
```

```
Client VM
anujtr@client-vm:~$ ping google.com
PING google.com (142.250.68.206) 56(84) bytes of data.
64 bytes from pnlgaa-ax-in-f14.1e100.net (142.250.68.206): icmp_seq=1 ttl=110 time=71.6 ms
64 bytes from pnlgaa-ax-in-f14.1e100.net (142.250.68.206): icmp_seq=2 ttl=110 time=81.5 ms
64 bytes from pnlgaa-ax-in-f14.1e100.net (142.250.68.206): icmp_seq=3 ttl=110 time=73.5 ms
^C
--- google.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 71.596/75.538/81.547/4.317 ms
anujtr@client-vm:~$ _
```

```
Server VM
anujtr@client-vm:~$ ping google.com
PING google.com (142.250.68.206) 56(84) bytes of data.
64 bytes from pnlgaa-ax-in-f14.1e100.net (142.250.68.206): icmp_seq=1 ttl=110 time=72.2 ms
64 bytes from pnlgaa-ax-in-f14.1e100.net (142.250.68.206): icmp_seq=2 ttl=110 time=81.9 ms
64 bytes from pnlgaa-ax-in-f14.1e100.net (142.250.68.206): icmp_seq=3 ttl=110 time=82.3 ms
^C
--- google.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2004ms
rtt min/avg/max/mdev = 72.238/78.813/82.325/4.653 ms
anujtr@client-vm:~$ _
```

```
Gateway VM
anujtr@client-vm:~$ ping google.com
PING google.com (142.250.68.206) 56(84) bytes of data.
64 bytes from pnlgaa-ax-in-f14.1e100.net (142.250.68.206): icmp_seq=1 ttl=111 time=71.8 ms
64 bytes from pnlgaa-ax-in-f14.1e100.net (142.250.68.206): icmp_seq=2 ttl=111 time=76.4 ms
64 bytes from pnlgaa-ax-in-f14.1e100.net (142.250.68.206): icmp_seq=3 ttl=111 time=80.0 ms
^C
--- google.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2008ms
rtt min/avg/max/mdev = 71.768/76.046/80.014/3.373 ms
anujtr@client-vm:~$
```


V. CONCLUSION

This project provided hands-on experience with virtual machine setup, UTM network adapter configuration, and Linux networking commands on Apple Silicon. A notable challenge was that the standard Ubuntu desktop AMD64 ISO is incompatible with the M1 Mac's ARM architecture, requiring the use of the Ubuntu 22.04.1 Server ARM64 ISO instead. Another challenge encountered was that cloned VMs shared the same MAC address, which caused network conflicts; this was resolved by clearing the MAC address field in UTM settings to auto-generate a new unique address for each VM. A key takeaway was the distinction between temporary network configuration (using `ip addr` and `ip route`, which are lost on reboot) and permanent configuration via netplan YAML files. Configuring the Gateway with three network interfaces and enabling IP forwarding and NAT masquerading illustrated how a router functions in a real network. One challenge encountered was ensuring that netplan YAML indentation was strictly correct, as any spacing errors would cause the configuration to fail silently. Another important lesson was the use of snapshots before making changes. The iptables MASQUERADE rule was made persistent using iptables-persistent to survive reboots. Overall, this project established a solid foundational virtual network that will be reused in subsequent projects for security and packet-analysis exercises.

VI. APPENDIX B: ATTACHED FILES

The following netplan configuration files were used.

Client VM - /etc/netplan/00-installer-config.yaml:

```
network:
  version: 2
  renderer: networkd
  ethernets:
    enp0s1:
      dhcp4: no
      addresses: [192.168.0.10/24] # Anuj T: Client IP on Net1
      gateway4: 192.168.0.100      # Anuj T: Gateway IP on Net1
      nameservers:
        addresses: [8.8.8.8, 8.8.4.4]
```

Server VM - /etc/netplan/00-installer-config.yaml:

```
network:
  version: 2
  renderer: networkd
  ethernets:
    enp0s1:
      dhcp4: no
      addresses: [10.0.0.10/8]      # Anuj T: Server IP on Net2
      gateway4: 10.0.0.100          # Anuj T: Gateway IP on Net2
      nameservers:
        addresses: [8.8.8.8, 8.8.4.4]
```

Gateway VM - /etc/netplan/00-installer-config.yaml:

```
network:
  version: 2
  renderer: networkd
  ethernets:
    enp0s2:
      dhcp4: no
      addresses: [192.168.0.100/24] # Anuj T: Gateway IP on Net1 (client-facing)
      nameservers:
        addresses: [8.8.8.8, 8.8.4.4]
    enp0s3:
      dhcp4: no
      addresses: [10.0.0.100/8]      # Anuj T: Gateway IP on Net2 (server-facing)
      nameservers:
        addresses: [8.8.8.8, 8.8.4.4]
    enp0s1:
      dhcp4: yes                     # Anuj T: Net3 DHCP via NatNetwork (Internet)
      nameservers:
        addresses: [8.8.8.8, 8.8.4.4]
```

VII. REFERENCES

- [43] UTM Virtual Machines for Mac. <https://mac.getutm.app/>
- [44] Canonical Ltd. "Ubuntu 22.04 LTS Server ARM64 Release."
<https://cdimage.ubuntu.com/releases/22.04.1/release/>
- [45] Ubuntu Desktop Download. <https://ubuntu.com/download/desktop>
- [46] Netplan Reference Documentation. <https://netplan.readthedocs.io/>

Video Link: <https://youtu.be/Olz7oePQMig>